# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Computer Science and Engineering

## CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

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| **Course Code:** | CSA10 / CSA1005 | **Course Title:** | Software Engineering |
| **CO Assessed:** | CO1 | **Assessment:** | Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop** |
| **Weightage:** | 20% | **Total Marks:** | 25 |
| **CO1 Statement:** | *Apply advanced methodologies and agile project management techniques to manage software projects effectively* | | |
| **Student Name:** | Patteela Yashwanth Kumar | **Reg. No.:** | 192471055 |
| **Date:** | 13-10-2026 | **Duration:** | 60 minutes |

**Code of Conduct**

*I Patteela Yashwanth kumar/192471055 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student:Patteela Yashwanth Kumar

## Annexure A – Agile Sprint Planning & User Story Workshop

**Instructions to Students:**

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders** ○ Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories** ○ Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>.*
3. **Define Acceptance Criteria** ○ Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog** ○ Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog** ○ Select high-priority user stories for the first sprint and divide them into actionable development tasks.

6. **Estimate Effort Using Story Points** ○ Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables** ○ Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

## Rubrics for Calculation

| Criteria | Excellent (5 Marks) | Good (3 Marks) | Satisfactory (2 Marks) | Needs Improvement (1 Mark) |
|---|---|---|---|---|
| **Project Vision & Stakeholder Identification (30%)** | Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified. | Vision is clear with minor omissions. Most stakeholders are correctly identified. | Vision is partially defined. Some stakeholders are missing or incorrectly identified. | Vision is unclear or incomplete. Stakeholders are poorly identified or missing. |
| **Epics & User Stories (25%)** | Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements. | Epics and user stories are mostly complete with minor formatting or requirement issues. | User stories are partially complete with limited Agile structure. | User stories are incomplete, unclear, or do not follow Agile practices. |
| **Acceptance Criteria (20%)** | Acceptance criteria are specific, measurable, testable, and aligned with all user stories. | Acceptance criteria are mostly clear and testable with minor gaps. | Acceptance criteria are partially defined and lack clarity or completeness. | Acceptance criteria are missing or not aligned with user stories. |
| **Product Backlog Prioritization & Sprint Planning (15%)** | Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete. | Backlog prioritization and sprint planning are mostly appropriate with minor issues. | Prioritization or sprint planning is partially complete with limited justification. | Backlog prioritization and sprint planning are poorly organized or incomplete. |
| **Story Point Estimation, Sprint Goal & Presentation (10%)** | Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized. | Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear. | Estimation or sprint goal lacks justification. Presentation is adequate but incomplete. | Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete. |

| Criteria | Mark |
|---|---|
| Project Vision & Stakeholder Identification **(30%)** | /5 |
| Epics & User Stories **(25%)** | /5 |
| Acceptance Criteria **(20%)** | /5 |
| Product Backlog Prioritization & Sprint Planning **(15%)** | /5 |

| | |
|---|---|
| Story Point Estimation, Sprint Goal & Presentation **(10%)** | /5 |
| **TOTAL** | /25 |

# CO1 ASSESSMENT TOOL 2

# AGILE SPRINT PLANNING & USER STORY WORKSHOP

## Smart Donation and Charity Management Platform

### 1. Project Vision and Stakeholders

**Vision:** To build a transparent, secure and easy-to-use platform that lets a charity manage donors, campaigns, donations, beneficiaries and resource distribution in one place, so that every donor can see how their contribution was used and can trust the organization. The vision is based on the problem that charities still keep scattered manual records, which reduces donor confidence and increases administrative work.

**Objectives:** register donors and create campaigns; record money and in-kind donations; issue automatic receipts; manage beneficiaries and track distribution; maintain financial records; and give donors dashboards and transparency reports.

**Success measures:** a donor can register and donate in under 3 minutes; every donation receives a receipt within 1 minute; and at least 95% of donations can be traced to a campaign and a distribution record.

**Key stakeholders and their roles:**

- **Donors:** individuals and institutions who donate, track fund usage and download receipts.
- **Charity Administrator:** creates campaigns, verifies donations and approves distribution.
- **Finance Team:** maintains the ledger, reconciles payments and supports audits.
- **Beneficiaries:** receive money, food, clothing or equipment and expect fair distribution.
- **Volunteers / Field Staff:** collect items and update delivery status.
- **Auditors / Regulators:** verify records and compliance.
- **Scrum Team:** the charity manager as Product Owner, a Scrum Master, developers and testers who plan, build and deliver the product.

### 2. Epics and User Stories

The requirements are grouped into 7 epics and 14 user stories, written in the format: As a <user>, I want <feature>, so that <benefit>.

**Epic E1 – Donor Management**

- US-01: As a donor, I want to register with my details, so that I can donate and track my contributions.
- US-02: As a user, I want to log in securely with my role, so that I access only the features permitted to me.

**Epic E2 – Campaign Management**

- US-03: As an admin, I want to create a campaign with a target and deadline, so that donors can contribute to a cause.
- US-04: As a donor, I want to view active campaigns, so that I can choose where to donate.

**Epic E3 – Donation and Receipts**

- US-05: As a donor, I want to donate online, so that my contribution is recorded instantly.
- US-06: As a donor, I want an automatic receipt, so that I have proof of my donation.
- US-07: As an admin, I want to record in-kind donations, so that goods are also tracked.

**Epic E4 – Beneficiary Management**

- US-08: As an admin, I want to register and verify beneficiaries, so that aid reaches genuine people.

**Epic E5 – Distribution and Inventory**

- US-09: As an admin, I want to allocate resources to beneficiaries, so that distribution is fair and recorded.
- US-10: As a volunteer, I want to update delivery status, so that progress is visible to everyone.

**Epic E6 – Finance Management**

- US-11: As a finance officer, I want a ledger of income and expenses, so that financial records are accurate.

**Epic E7 – Dashboards and Reports**

- US-12: As a donor, I want a dashboard of my donations and how they were used, so that I trust the charity.
- US-13: As an admin, I want to generate transparency reports, so that I can share fund usage with the public.
- US-14: As an admin, I want to view campaign performance, so that I can plan future campaigns better.

The stories follow the INVEST guidelines: each is independent, valuable to a user, small enough for one sprint and testable. Epics such as Donor Management and Campaign Management are large features that are split into these smaller stories.

**3. Acceptance Criteria (Given–When–Then)**

Acceptance criteria are written for the six user stories selected for Sprint 1. Each one is specific, measurable and testable.

**US-01 Donor registration**

- Given the registration page, when I enter a valid name, email, phone and password, then my account is created and a confirmation email is sent within 1 minute.

- Given an email that is already registered, when I submit it again, then an error message is shown and no duplicate account is created.

**US-02 Secure login**

- Given valid credentials, when I log in, then I reach the dashboard of my role within 3 seconds.
- Given 5 wrong attempts, when I try again, then the account is locked for 15 minutes.

**US-03 Create campaign**

- Given an admin is logged in, when a campaign with title, target amount and end date is saved, then it appears in the active campaign list.
- Given an end date in the past, when I save, then the system rejects it with a clear message.

**US-04 View campaigns**

- Given active campaigns exist, when a donor opens the campaigns page, then each campaign shows title, target, amount raised and days left.

**US-05 Online donation**

- Given a campaign and an amount, when the payment succeeds, then the donation is stored and the campaign total updates immediately.
- Given a failed payment, when the gateway returns an error, then no donation is recorded and the donor sees a retry option.

**Common criteria for all stories:** pages load within 3 seconds, all data is sent over HTTPS, passwords are stored in hashed form, and payment details are never stored in the application database.

**US-06 Automatic receipt**

- Given a successful donation, when it is saved, then a PDF receipt with a unique number, donor name, amount and date is emailed within 1 minute.
- Given the donor dashboard, when I open my donations, then I can download the receipt again.

**4. Prioritized Product Backlog**

The product backlog is ordered using three factors: business value (how much trust and funding the feature brings), donor and charity needs, and dependencies (a feature that others rely on comes first). The MoSCoW technique is used: M = Must have, S = Should have, C = Could have.

| Rank | Story | Priority | Reason / dependency |
|---|---|---|---|
| 1 | US-01 Donor registration | M | Base for donor features |
| 2 | US-02 Secure login | M | Needed for protected features |
| 3 | US-03 Create campaign | M | Donations need campaigns |
| 4 | US-04 View campaigns | M | Entry point for donors |
| 5 | US-05 Online donation | M | Core business value |

| Rank | Story | Priority | Reason / dependency |
|---|---|---|---|
| 6 | US-06 Automatic receipt | M | Depends on US-05 |
| 7 | US-08 Register beneficiary | M | Needed for distribution |
| 8 | US-09 Allocate resources | M | Depends on US-07, US-08 |
| 9 | US-07 In-kind donation | S | Extends donation types |
| 10 | US-10 Delivery status | S | Depends on US-09 |
| 11 | US-11 Finance ledger | S | Depends on donations |
| 12 | US-12 Donor dashboard | S | Needs stored data |
| 13 | US-13 Transparency report | S | Needs ledger and distribution |
| 14 | US-14 Campaign performance | C | Analytics; can follow later |

In short, donor registration, login, campaigns, donation and receipts are built first because they create the money flow and give early value. Beneficiary and distribution features follow, and reporting and analytics come last because they depend on the data created by the earlier stories.

**5. Sprint Backlog (Sprint 1 – 2 weeks)**

Stories selected for Sprint 1: US-01, US-02, US-03, US-04, US-05 and US-06, a total of 26 story points. The team has 4 members working 8 effective days at 6 focus hours per day, which gives a capacity of 192 hours. The tasks below total 164 hours (about 85% of capacity), leaving a buffer for defects and rework.

- **US-01 (22 h):** design donor table (4 h), build registration form and validation (8 h), email confirmation service (6 h), unit and acceptance tests (4 h).
- **US-02 (26 h):** authentication and password hashing (8 h), role-based access control (8 h), account lockout rule (4 h), security test cases (6 h).
- **US-03 (24 h):** campaign database and API (8 h), admin campaign form (8 h), date and amount validation (4 h), campaign tests (4 h).
- **US-04 (14 h):** campaign list page (6 h), progress bar and days-left display (4 h), hide closed campaigns and test (4 h).
- **US-05 (48 h):** payment gateway sandbox integration (16 h), donation API (10 h), campaign total update (6 h), payment failure handling (6 h), payment testing (10 h).
- **US-06 (30 h):** receipt template (6 h), unique number and PDF generation (10 h), email delivery (8 h), download option and testing (6 h).

**6. Story Point Estimation**

**Technique:** Planning Poker with the Fibonacci sequence (1, 2, 3, 5, 8, 13). Each team member privately selects a card, the cards are revealed together, and differences are discussed until the team agrees. US-02 (login) was taken as the reference story with 3 points, and other stories were compared

with it for complexity, effort and risk. For example, the first votes for US-05 were 5, 8, 8 and 13; after the developer with the highest vote explained the payment-gateway risk, the team agreed on 8 points.

| Story | Points | Justification |
|---|---|---|
| US-01 | 3 | Standard form, validation and email; low risk |
| US-02 | 3 | Reference story; needs security testing |
| US-03 | 5 | New data model, admin form and validations |
| US-04 | 2 | Read-only display of campaign data |
| US-05 | 8 | External payment integration; highest risk |
| US-06 | 5 | PDF generation and email with unique numbers |

The total of 26 points matches the assumed velocity for one 2-week sprint, so the sprint is realistic.

**7. Sprint Goal and Expected Deliverables**

**Sprint Goal:** "Allow a donor to register, log in, view active campaigns and make an online donation that is recorded and acknowledged with an automatic receipt."

**Expected deliverables at the end of Sprint 1:**

- Working donor registration and secure role-based login.
- Admin module to create and manage campaigns.
- Campaign listing page for donors with live progress.
- Online donation with payment integration (sandbox) and updated campaign totals.
- Automatic PDF receipts sent by email and downloadable.
- Test report, updated product backlog and a demo for stakeholders.

**Definition of Ready:** a story enters the sprint only when it has a clear statement, acceptance criteria, an agreed estimate and no unresolved dependency.

**Sprint plan:** Day 1 is Sprint Planning, when tasks are assigned and the Definition of Done is agreed. Days 2 to 9 are development and testing with a 15-minute Daily Stand-up. Day 10 holds the Sprint Review, where the team demonstrates the work to stakeholders, and the Sprint Retrospective, where the team agrees one improvement for Sprint 2. The planned burn-down is 26 points on Day 1, 21 on Day 3, 14 on Day 5, 6 on Day 8 and 0 on Day 10.

**Risks and mitigation:** payment gateway delays are handled by starting US-05 early and using the sandbox; security flaws in login are reduced by peer code review and security tests; scope creep is avoided by freezing the sprint backlog and moving new ideas to the product backlog; and an absent team member is covered by pair programming.

**Definition of Done:** code reviewed and merged, unit and acceptance tests passed, no critical defects, deployed to the test server and accepted by the Product Owner in the Sprint Review.

**Sprint Review and Retrospective:** In the Sprint Review the team demonstrates registration, login, campaigns, donation and receipt to the stakeholders and records their feedback in the product backlog. In the Retrospective the team discusses what went well, what did not go well, and one improvement action for the next sprint.

**Why Fibonacci numbers:** the gaps between 1, 2, 3, 5, 8 and 13 grow larger, which shows that bigger stories are less certain and should be split or discussed further before being committed to a sprint.

**Next sprint:** Sprint 2 will cover US-07, US-08, US-09 and US-10 (in-kind donations, beneficiary registration, resource allocation and delivery status), and Sprint 3 will cover finance, dashboard and reporting.




